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Credit Risk Prediction: Balancing Financial Inclusion and Risk Mitigation

Building a Classification Model: Logistic Regression vs LightGBM for Credit Decision Optimization

Presented By:

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Background

PT Home Credit Indonesia adalah **perusahaan PT Home Credit Indonesia is a multinational consumer finance company** serving more than 3.4 million customers through both online and offline instant financing services. With the **massive volume of credit applications processed daily**, **the company's conventional credit evaluation system is prone to producing False Positives incorrectly rejecting potential customers who are actually eligible for loans**. This condition puts the company at risk of losing significant revenue opportunities and limits its ability to optimize loan parameters effectively to ensure successful repayment.

Research Problem

The conventional credit evaluation system is prone to generating False Positives, leading to the rejection of potential customers and lacking the ability to optimize loan parameters (principal, tenor, repayment schedule) to improve repayment success.

Business Understanding & Project Objectives

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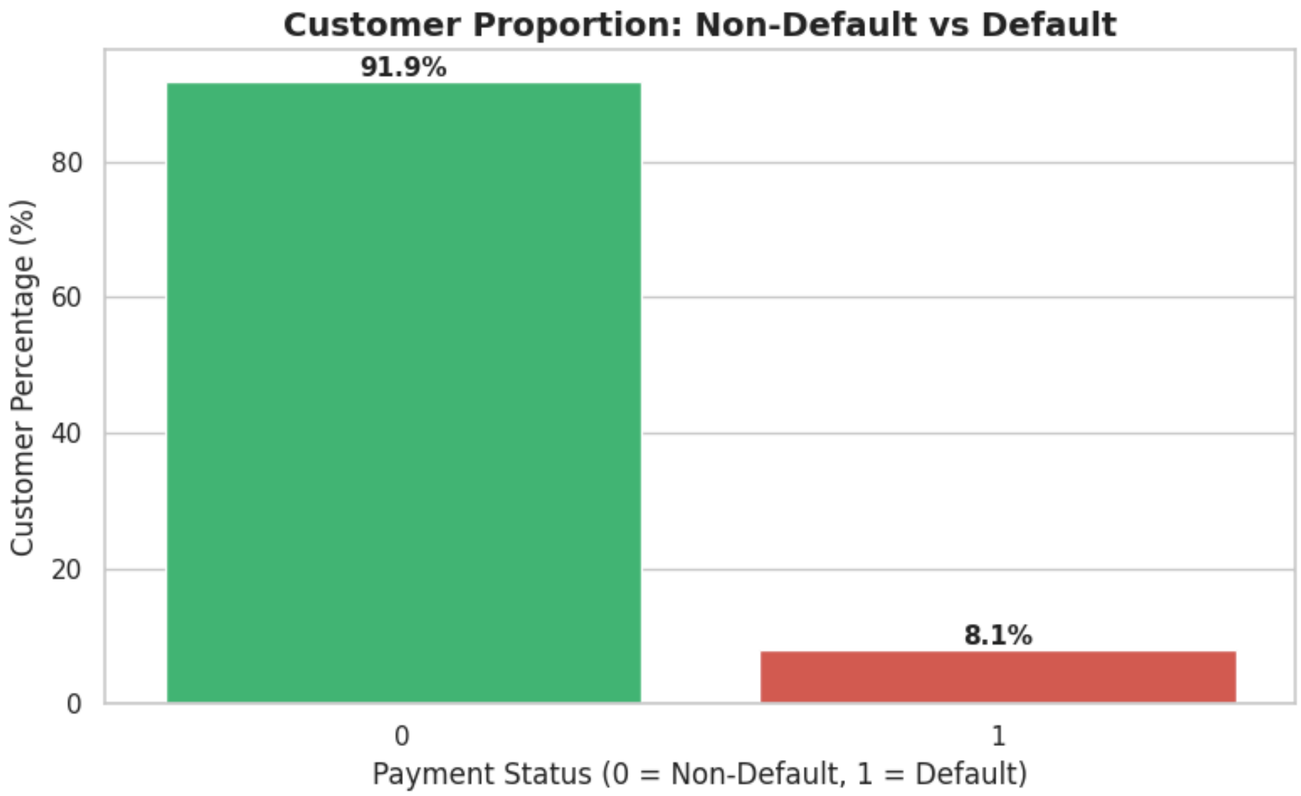


Data Characteristics:

The class distribution in the historical dataset **is highly imbalanced**, dominated by Target 0 / **Non-Default at 92%** (Majority Class), compared to Target 1 / **Default at 8%** (Minority Class).

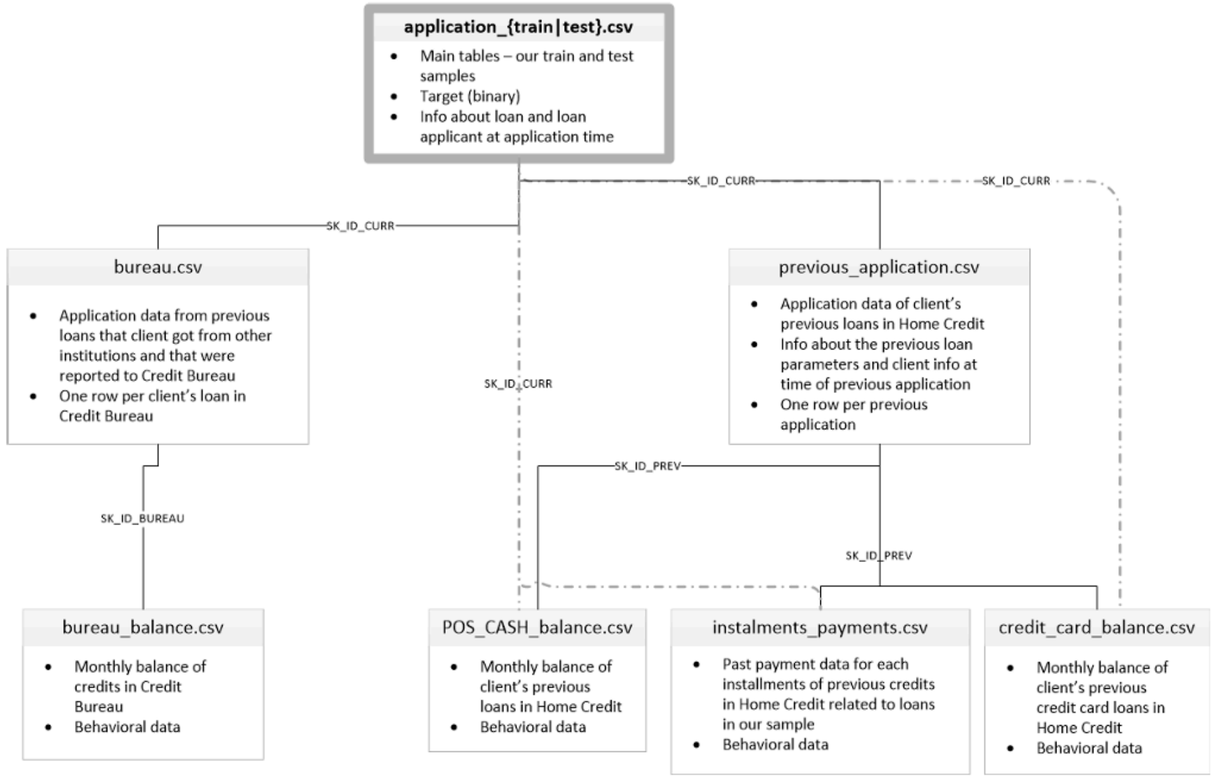
Objectives:

- **Develop and compare Machine Learning classification models** to accurately **identify customer risk**.
- **Reduce False Positives**, i.e., rejecting customers who are actually creditworthy.
- **Leverage model prediction probabilities** to design personalized loan parameters (principal, tenor, installments).
- **Improve repayment success without increasing** Non-Performing Loans (**NPL**).

Evaluation Metrics:

- **Accuracy is not used** as it is not representative for imbalanced datasets.
- **Class weighting is applied** (e.g., is_unbalance=True in LightGBM) to assign a higher penalty to the minority class. This approach makes the model more sensitive in detecting high-risk customers (increasing Recall).
- **To prevent excessive False Positives** (rejecting good customers), **evaluation focuses on**:
  - **PR-AUC** → Maintains the trade-off between Recall and Precision for the minority class.
  - **ROC-AUC** → Evaluates the model's ability to distinguish between different levels of customer risk.

Data Summary



1. **Main Table (Core Data)**
   - **File**: application_{train|test}.csv.
   - **Description**: The main table that contains samples of train and test data. The table includes information about loans and customers at the time of application, and also contains a binary Target variable.
   - **Relationship Key**: Uses SK_ID_CURR to connect with other tables.

2. **Internal Data (Customer History)**
   - **Main File**: previous_application.csv.
   - **Description**: Records information on parameters and customer application data from previous loans within the company.
   - **Monthly Behavioral Data**: Details of historical records from past company loans are linked via SK_ID_PREV to three detailed tables:
     - **POS_CASH_balance.csv**: Monthly balance data for cash loans and point of sales (POS).
     - **credit_card_balance.csv**: Monthly balance data of customer credit cards.
     - **installments_payments.csv**: Historical payment data for each loan installment.

3. **External Data (Credit Bureau History)**
   - **Files**: bureau.csv and bureau_balance.csv.
   - **Description**: Records customer loan data from other financial institutions reported to the Credit Bureau. This information is complemented with behavioral data in the form of monthly loan balances.

Data Pre-Processing

1. **Removal of data with >60% missing parameters**

Removing 17 columns (mostly related to building/residential profiles) that have more than 60% missing values to prevent noise in the model.

2. **Removal of Duplicate Values**

Validating data integrity (no duplicate rows found based on Customer ID/SK_ID_CURR).

3. **Data Consistency & Anomaly Handling**

- Converting time format (DAYS_ columns) into absolute (positive) values.
- Replacing unreasonable values (such as employment duration of 365243 days (>100 years)) with 0, assuming unemployed/retired status.
- Replacing ambiguous text values (such as 'XNA', 'not specified') with N/A.

4. **Imputation Strategy (Filling N/A)**

- Categorical Data (Text): Generally filled with a new category 'Unknown'. (Special case: anomaly in Gender column is filled with its mode value, 'F').
- Specific Numerical Data (DAYS_EMPLOYED): Filled with 0 (assumed to represent unemployed/retired status).
- General Numerical Data: Filled using the median value. Additionally, a special indicator column (_IS_MISSING flag) is created so the model can still recognize that the data was originally missing.

Data Transform

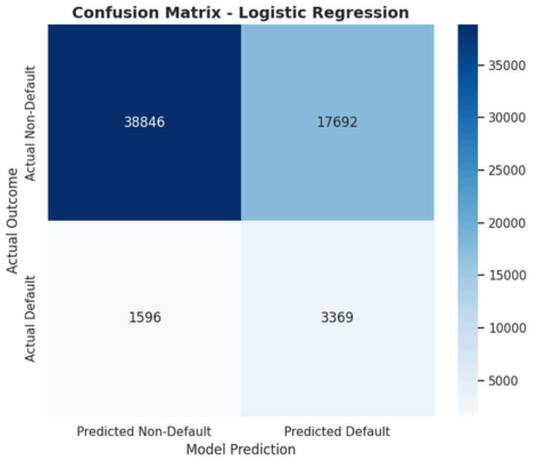
1. **Data Splitting (80:20)**

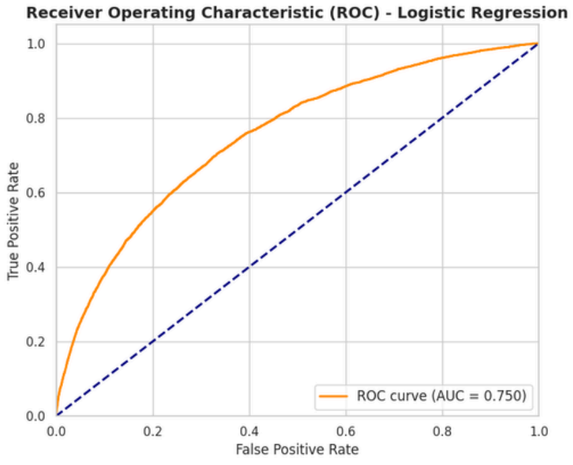
Splitting application_train, which contains complete TARGET column data, into training and testing sets using Stratified Sampling to ensure the imbalance ratio (8% default) remains consistent in both datasets.

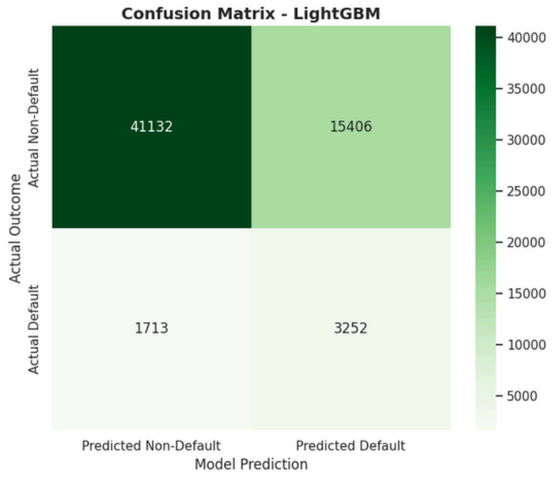
2. **Feature Scaling**

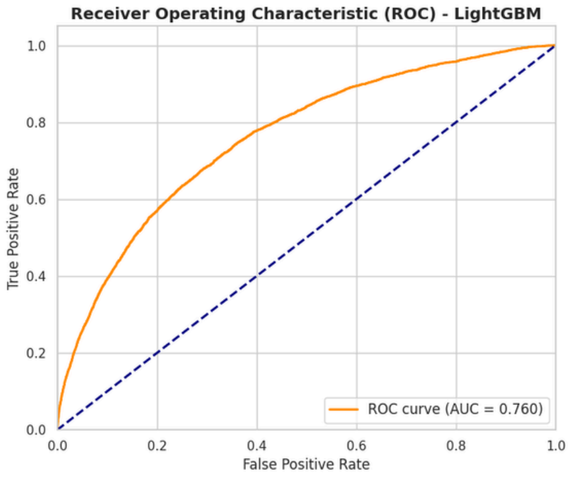
Standardizing the scale range of all numerical features (Standardization) so that linear models are not biased toward large numerical values, applied carefully to prevent data leakage into the test set.

Machine Learning Implementation and Evaluation







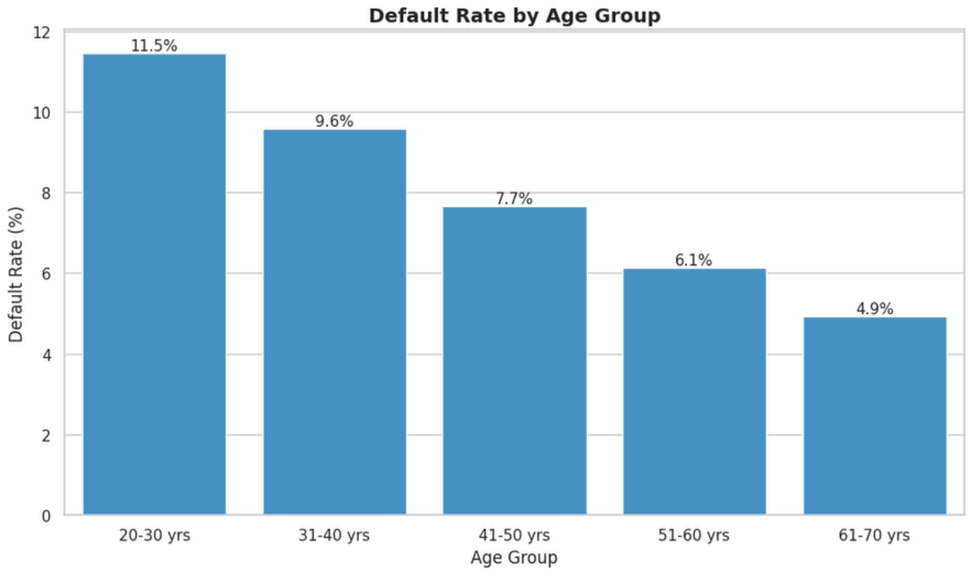


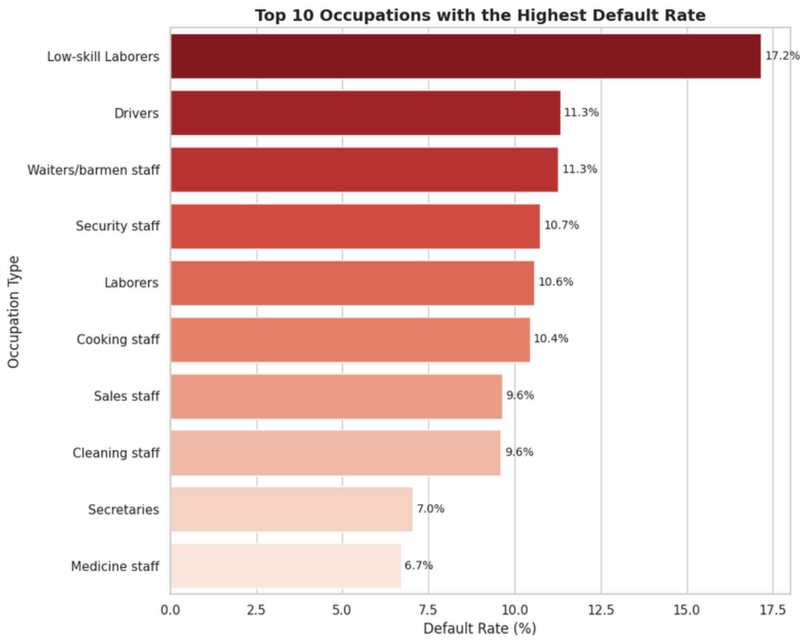
- Detection Capability (True Positives): Successfully predicted 3,369 / 4,965 (68%) customers who actually defaulted.
- False Rejections (False Positives): Produced 17,692 default predictions for customers who were actually non-default.
- ROC-AUC = 0.750, indicating the model's basic ability to distinguish customer risk.

- Detection Capability (True Positives): Successfully predicted 3,252 / 4,965 (65%) customers who actually defaulted.
- False Rejections (False Positives): Produced 15,406 default predictions for customers who were actually non-default.
- ROC-AUC = 0.760, indicating good model capability in distinguishing customer risk.

Therefore, although Logistic Regression is slightly more sensitive in detecting risk, **LightGBM is more recommended**. **This model trades a small risk of 117 additional default cases** (False Negatives) **in order to save 2,286 potential non-default customers** (False Positives) from incorrect rejection. From a business perspective, this trade-off is far more beneficial for increasing revenue.

Data Visualization and Business Insight

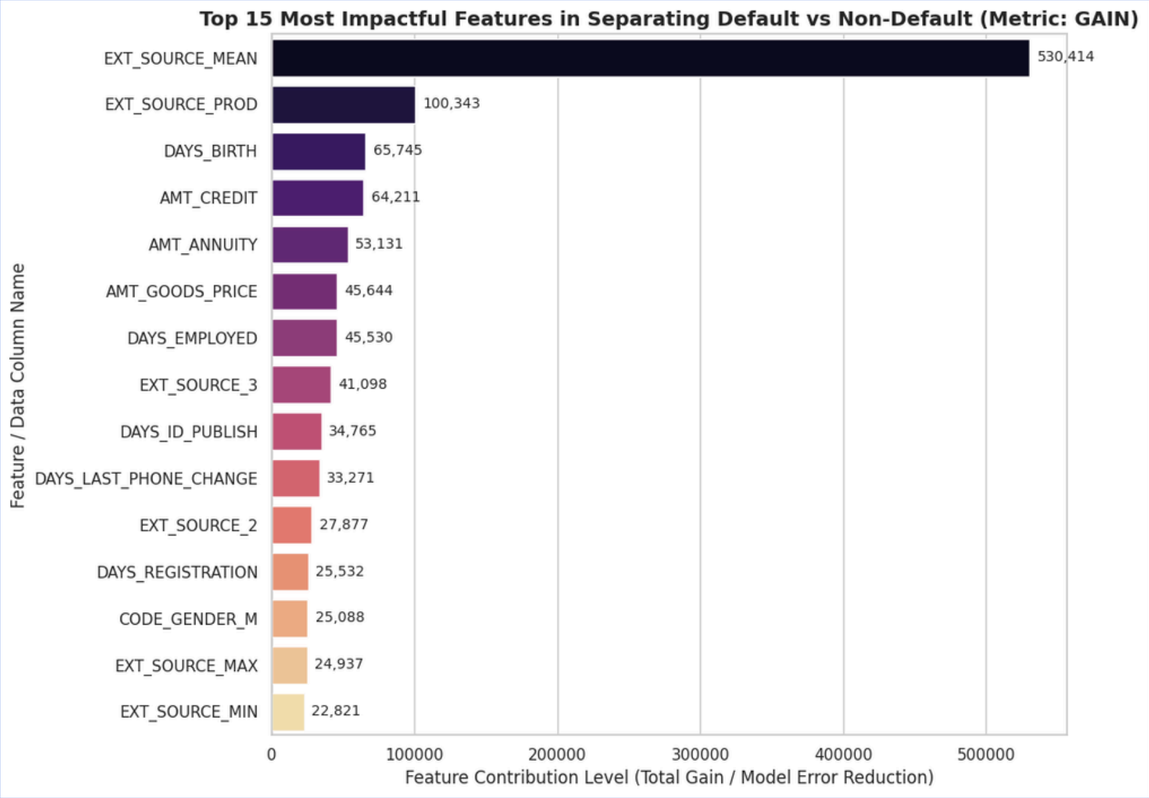




Insight Data

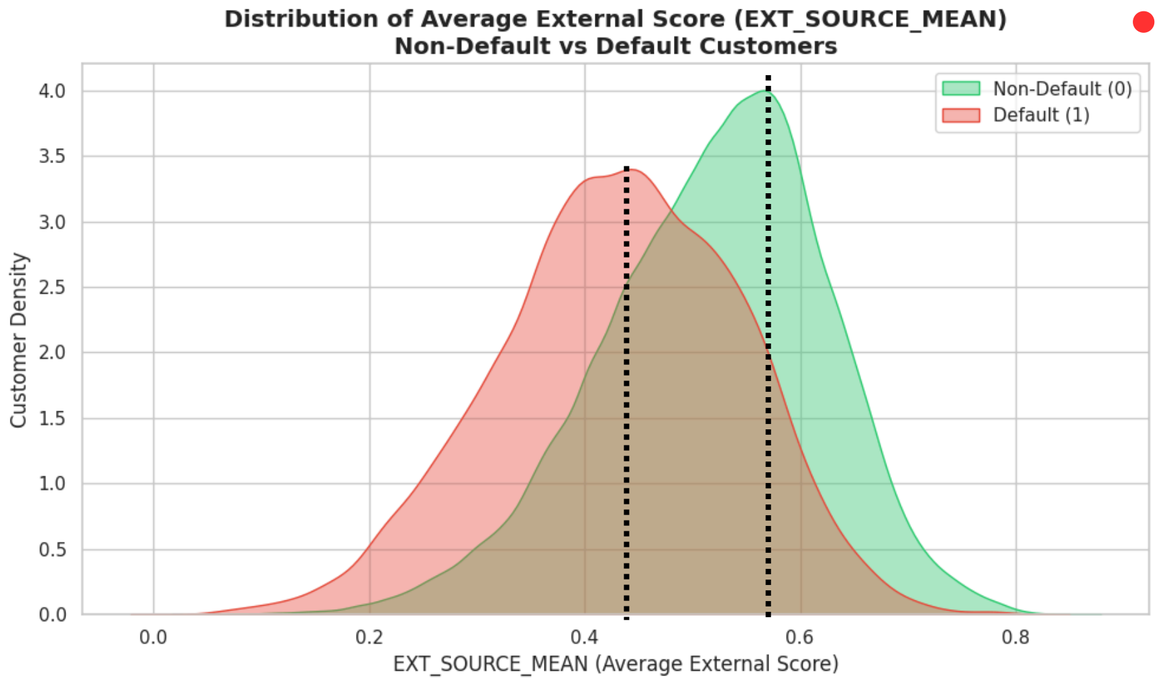
- Customers aged 20–30 have the highest default risk (±11.5%), which consistently decreases with age (61–70 years: ±4.9%).
- High-risk occupations: Low-skill laborers (17.2%), drivers (11.3%), and waiters/barmen (11.3%) → vulnerable due to unstable cash flow.
- Recommendation 1 (Risk Control): Tighten underwriting or require additional collateral for young customers with high-risk occupations.
- Recommendation 2 (Product Strategy):
  - Implement product customization → start with small loan limits & short tenors (micro-loans)
  - Gradually increase limits if repayment history is good
  - Use external scores (EXT_SOURCE) as the main filter to ensure fairness for potential customers.

Feature Importance



Insight Data

- GAIN Metric Approach: **Using the Gain evaluation metric** (Impact of Error Reduction) **to address bias in imbalanced datasets**.
- **Third-Party Validation** as Key: **Scores from external credit bureaus** (EXT_SOURCE_MEAN), which represent the average track record or 'report card' of customer loans from other financial institutions, **are proven to be the most impactful predictor** compared to basic factors such as age or income.



- **Risk Threshold**: **The peak density of default customers** (red area) **is concentrated in the lower score range** (peaking around 0.43). In contrast, the **non-default population** (green area) shifts to the right and **dominates higher score ranges** (peaking around 0.57).

Conclusion:

Based on the data, the external credit report score (EXT_SOURCE_MEAN) is the most accurate risk predictor: **the lower the external score, the more likely the customer is to default.**

Business Recommendation

- **Optimization of Main Filter**: **Use external credit bureau scores** (EXT_SOURCE_MEAN) **as the primary determinant of credit approval**, replacing the dominance of profile factors such as age and income.
- **Product Customization & Tiering**: **Provide small initial loans (micro-loans) with short tenors, then gradually increase limits** if repayment history is disciplined **for at-risk customers** (aged 20-30 years & blue-collar workers).
- **Revenue Maximization** (LightGBM): **Implementing the LightGBM** model **has been proven to reduce False Positives better than Logistic Regression**, minimizing system errors in rejecting thousands of potential customers.

Future Works

- **Behavioral Data Extraction**: **Processing secondary tables** (installments_payments.csv) **to obtain features of past installment behavior**, such as average Days Past Due.
- **Model Segmentation** (Clustering): **Building specific models separated based on customer profiles** (e.g., first-time borrowers vs experienced customers) to ensure fairer assessment.
- **Advanced Ensembling**: **Combining the strengths of multiple algorithms** (LightGBM, XGBoost, Neural Network) using the Stacking method **to maximize accuracy**.

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Thank You Very Much

The complete application can be accessed at:
https://github.com/niqanabila16/Credit-Score-Prediction